Trader Behavior vs Market Sentiment Analysis – Charts, Tables And Inference

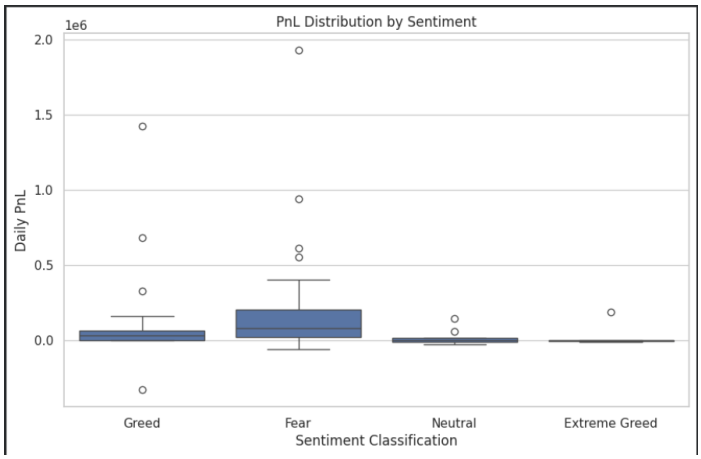
1. Performance vs Sentiment

Figure 1: Summary Table

	daily_pnl			win_rate			drawdown_proxy		
	mean	median	std	mean	median	std	mean	median	std
classification									
Extreme Greed	35393.0984	0.0000	85325.4999	33.6609	30.6849	30.2367	-2696.6974	0.0000	3785.3471
Fear	209372.6622	81389.6825	380423.9069	41.5878	39.3962	13.6967	-2496.1294	0.0000	10990.2283
Greed	99675.5167	35988.3764	282694.2855	37.4074	41.2599	27.7185	-10234.5594	0.0000	57895.4107
Neutral	19842.7973	-0.4186	57227.7879	26.0683	18.8324	30.7062	-6202.8962	-0.5000	9720.5910

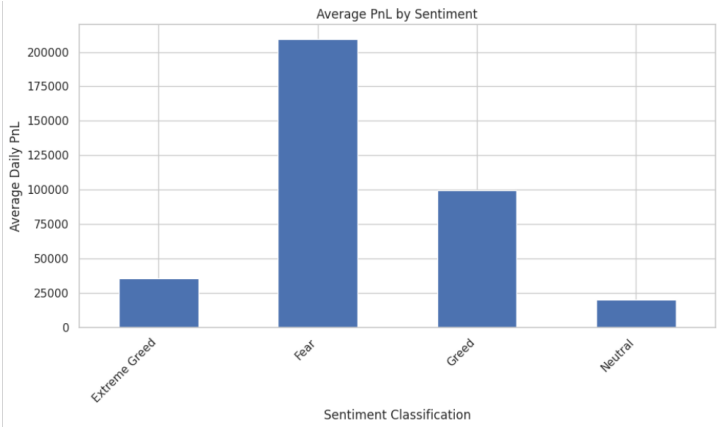
Insight: Fear shows highest average PnL but also high variance, indicating unstable gains.

Figure 2: PnL Distribution



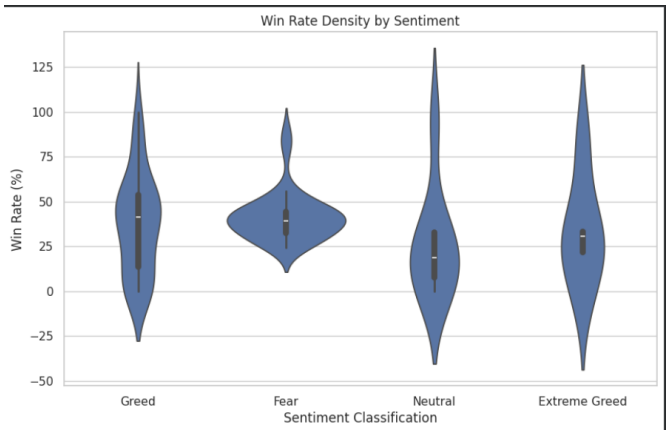
Insight: Fear and Greed exhibit wider spreads and extreme outliers, implying higher risk.

Figure 3: Average PnL



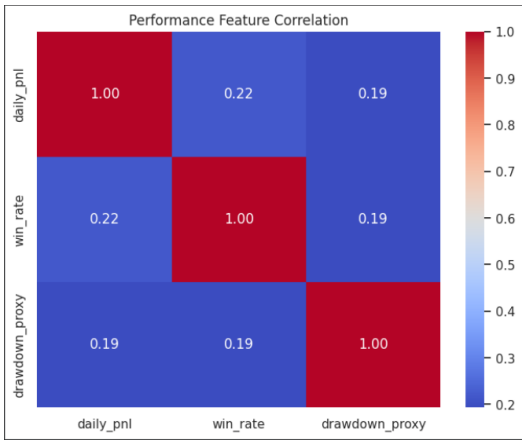
Insight: Fear dominates mean returns, while Neutral shows weakest performance.

Figure 4: Win Rate Distribution



Insight: Win rates are inconsistent across sentiments, with no strong stability advantage.

Figure 5: Correlation Heatmap



Insight: Weak correlations suggest profitability is not driven by a single factor.

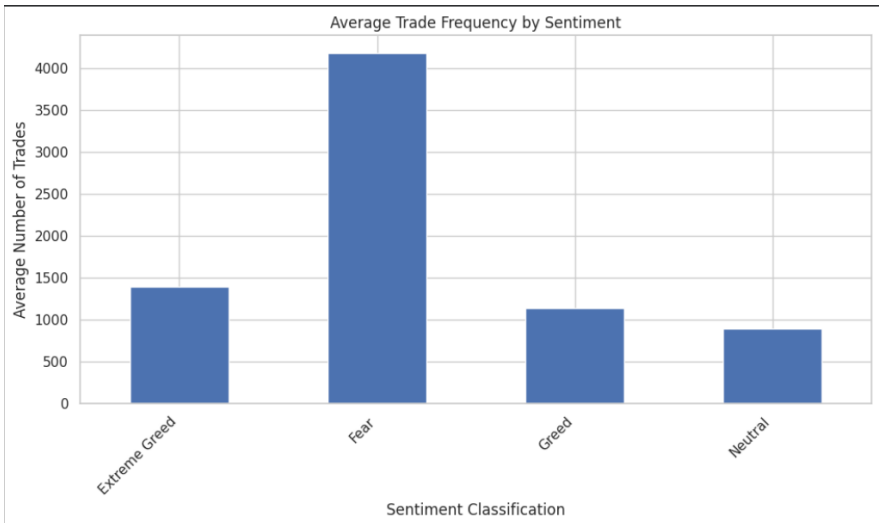
2. Behavior vs Sentiment

Figure 6: Behavior Summary Table

	num_trades			avg_trade_size_usd			long_short_ratio			risk_score		
	mean	median	std	mean	median	std	mean	median	std	mean	median	std
classification												
Extreme Greed	1392.4000	730.0000	1545.7151	4344.4478	3677.8091	2588.0283	inf	0.9571	NaN	7881354.0500	4540452.9000	9840995.0862
Fear	4183.4688	2763.5000	5730.6754	5926.5227	3207.8879	6881.6933	0.9681	0.8916	0.5183	22004952.9056	7341059.7850	36424885.9078
Greed	1134.0312	275.0000	2139.8096	5839.3110	2709.5322	7493.6332	inf	0.8818	NaN	3609489.7453	803879.2450	7409434.7394
Neutral	892.6250	164.5000	1342.9593	3793.4442	2317.5857	3782.8787	1.2433	0.9054	1.1584	2730404.2938	372722.0800	4622279.5660

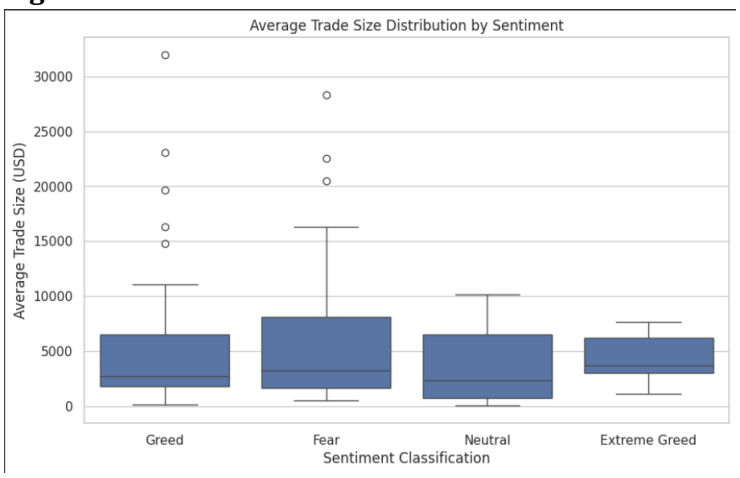
Insight: Fear drives highest trading activity and risk exposure.

Figure 7: Trade Frequency



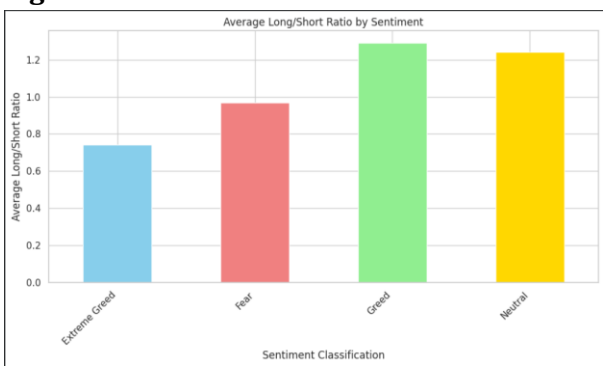
Insight: Traders are most active during Fear, confirming emotion-driven trading.

Figure 8: Position Size



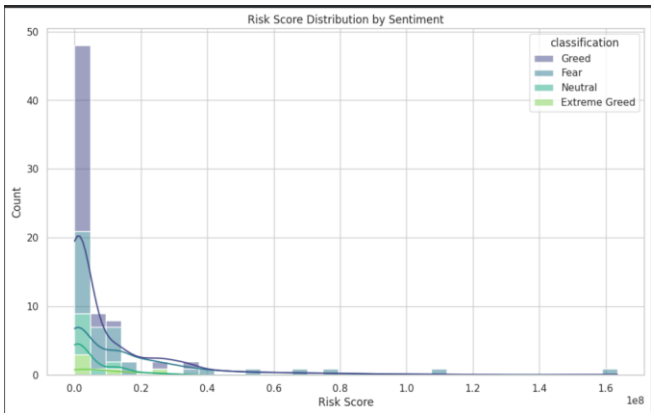
Insight: Larger trades occur during Fear and Greed, indicating increased conviction.

Figure 9: Directional Bias



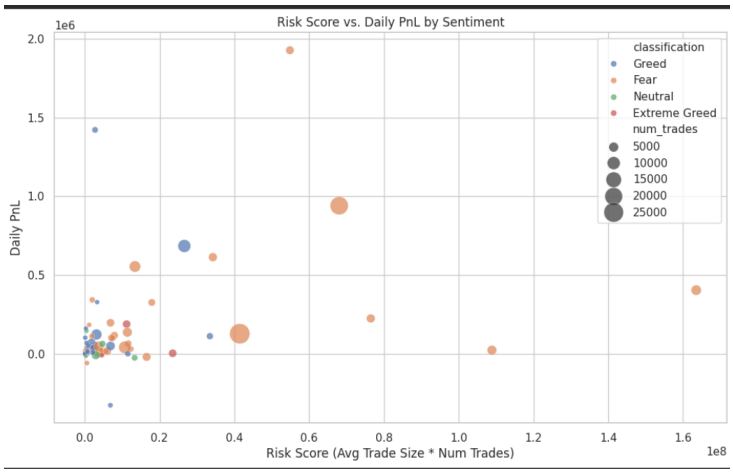
Insight: Greed shows long bias, while Fear is more balanced.

Figure 10: Risk Distribution



Insight: Risk is heavily skewed with extreme outliers in Fear and Greed.

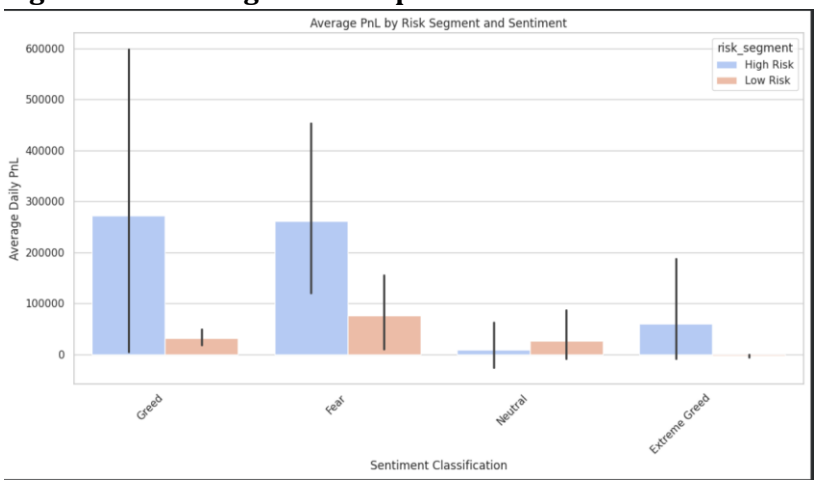
Figure 11: Multivariate Scatter



Insight: Higher risk generally leads to higher PnL variability.

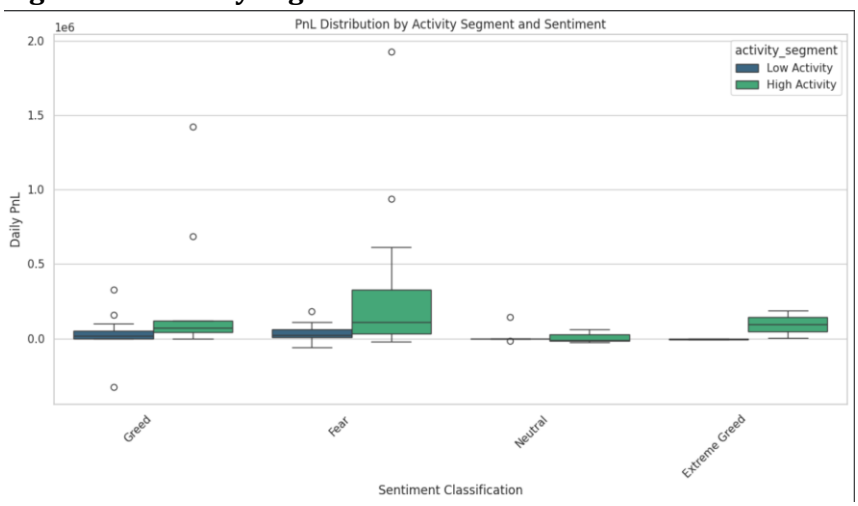
3. Segment Analysis

Figure 12: Risk Segment Comparison



Insight: High-risk traders achieve higher returns but with extreme variability.

Figure 13: Activity Segment Distribution



Insight: High activity traders dominate extreme outcomes.

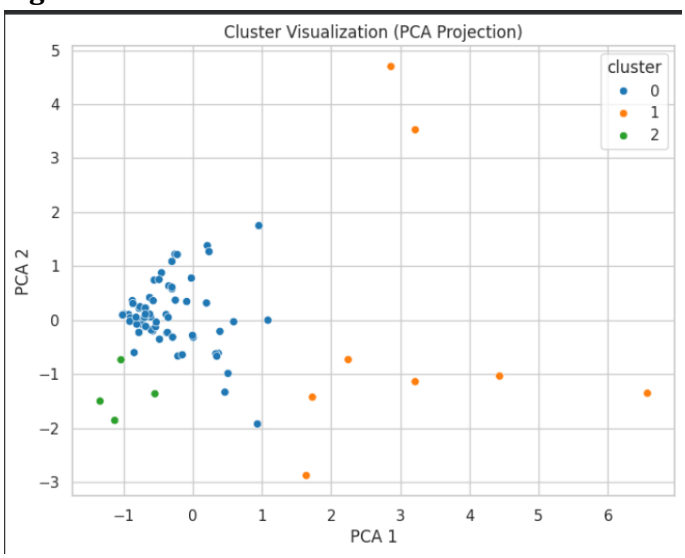
4. Clustering (Behavioral Archetypes)

Figure 14: Cluster Profile Table

cluster	num_trades	avg_trade_size_usd	risk_score	long_short_ratio	daily_pnl	win_rate
0	1787.3846	4169.1038	5101319.3212	0.8271	99586.9219	38.5840
1	8407.1250	17996.8732	68353278.2850	1.0978	469730.3859	45.4021
2	206.5000	3402.4350	625046.8175	5.4796	-1435.8539	8.3807

Insight: Distinct groups emerge: balanced, aggressive, and conservative traders.

Figure 15: PCA Cluster Visualization



Insight: Aggressive traders form a clearly separable group in feature space.